*Last updated in October 2026*

# Zekun Wang

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## Research Focus

I study continual learning for deployment-time adaptation and generalization in foundation models and AI agents. I investigate how these systems can learn from ongoing experience, adapt to unfamiliar tasks and environments, and improve over time while preserving prior knowledge and capabilities.

## Education

**Georgia Institute of Technology** *2024 – 2029*
*Ph.D. in Computer Science*
Advisor: Christopher J. MacLellan

**University of Michigan** *2022 – 2024*
*M.S. in Computer Science*

**Pennsylvania State University** *2018 – 2022*
*B.S. in Computer Science*
Double major: Mathematics. Minor: Economics.

## Industry Experience

**Amazon** *08/2026 – present*
*Research Intern · WA, USA*

- **One-shot adaptation to novel tasks.** Developing methods that adapt an agent to an unfamiliar task at deployment through a single update to its harness.
- **Self-improving agent harnesses.** Contributing to a harness self-improvement system that outperforms baseline harness-evolution methods on SWE-bench and Terminal-Bench.

## Selected Publications

For the full publication list, see Google Scholar. * Equal contribution.

**Trust Region Continual Learning as an Implicit Meta-Learner** *NeurIPS 2026*
**Zekun Wang***, Anant Gupta*, Christopher J. MacLellan

- Proved that replay with a Fisher-metric trust region yields a one-step MAML-like update under local approximations, connecting continual learning to meta-learning.
- Improved retention and re-convergence over EWC, replay, FTML, and VR-MCL on ImageNet-500 diffusion generation and Continual-World-10 diffusion policies.

**Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion Models** *ICLR 2026*
**Zekun Wang***, Anant Gupta*, Zihan Dong, Christopher J. MacLellan

- Derived and empirically validated a low-SNR rank-1 Fisher approximation near convergence, enabling structured EWC with generative replay.
- Reduced mean FID-based forgetting by **67% on ImageNet-1k** (32×32, 20 tasks) versus generative distillation alone; nearly eliminated forgetting on MNIST and FashionMNIST.

**Deep Taxonomic Networks for Unsupervised Hierarchical Prototype Discovery** *NeurIPS 2025*
**Zekun Wang**, Ethan Haarer, Tianyi Zhu, Zhiyi Dai, Christopher J. MacLellan
- Developed a hierarchical generative model that discovers prototypes without a known class count and achieved state-of-the-art hierarchical prototype discovery in the paper's evaluation.
- Proved that its variational objective also optimizes mutual information between latent representations and the hierarchy.

### Selected Preprints

**Self-Consolidating Language Models** *Preprint*
**Zekun Wang**[*], Anant Gupta[*], Zihan Dong, Christopher J. MacLellan
- Developed a meta-RL post-training framework in which an LLM learns which layers to update as new context arrives, balancing knowledge acquisition against forgetting.
- Improved SQuAD acquisition and retention over sequential SEAL, and LongBench v2 accuracy over sequential fine-tuning; learned update policies transferred to longer context streams.

**Test-Time Compositional Generalization in Diffusion Models via Concept Discovery** *Preprint*
**Zekun Wang**, Anant Gupta, Tianyi Zhu, Christopher J. MacLellan
- Developed test-time concept discovery and composition from a pretrained diffusion model's scores, enabling novel combinations **without retraining the base model**.
- Outperformed query-only and nearest-class baselines on ColorMNIST and CelebA composition using a product-of-experts sampler and a distilled variant.

## Honors and Awards

**Ada Lovelace Award for Excellent Paper** *2026*
*Advances in Cognitive Systems (ACS)*
For Diffusion Models and Concept Formation.

## Patent Applications

**Systems and Methods for Continual Learning of Diffusion Models**
U.S. Patent Application No. 19/821,778, filed September 28, 2026.
*Co-inventor. Applicant: Georgia Tech Research Corporation.*

**System and Method for Deep Taxonomic Networks for Unsupervised Hierarchical Prototype Discovery and Classification**
U.S. Patent Application No. 19/822,642, filed September 28, 2026.
*Co-inventor. Applicant: Georgia Tech Research Corporation.*

**Method and Apparatus for Taxonomic Networks with Neuro-Symbolic Pairing**
U.S. Patent Application No. 19/694,309, filed June 1, 2026.
*Co-inventor. Applicant: Georgia Tech Research Corporation.*

**Information-Theoretic Hierarchical Concept Learning for Masked-Word Prediction**
U.S. Patent Application No. 19/237,713, filed June 13, 2025.
*Co-inventor. Applicant: Georgia Tech Research Corporation.*

## Research Experiences

**Georgia Institute of Technology** *Aug. 2024 – present*
*Graduate Research Assistant — Teachable AI Lab*

**University of Michigan** *2022 – 2024*
*Graduate Research Assistant — SLED Lab (PI: Joyce Chai)*

**Pennsylvania State University** *2021 – 2022*
*Undergraduate Research Assistant*

## Teaching Experiences

**Graduate Student Instructor — EECS 595, Natural Language Processing** *Sept. 2023 – May 2024*
*University of Michigan*

**Teaching Assistant — CMPSC 464, Introduction to the Theory of Computation** *Jan. 2022 – May 2022*
*Pennsylvania State University*

**Teaching Assistant — MATH 230, Calculus and Vector Analysis** *Aug. 2021 – Dec. 2021*
*Pennsylvania State University*

**Math Peer Tutor — Calculus** *Jan. 2020 – Jan. 2022*
*Penn State Learning*

## Invited Talks and Guest Lectures

**Intro to AI (CS 3600), Georgia Tech** — Guest lecture *Aug. 2025*

**Deep Learning: Classics and Trends (DLCT)** — Invited talk *July 2024*
*Babysit a Language Model From Scratch: Interactive Language Learning by Trials and Demonstrations*

**Advanced Artificial Intelligence (EECS 692), U. of Michigan** — Guest lecture *Jan. 2023*

## Academic Leadership and Service

**Workshop Organizer** *Dec. 2026 (upcoming)*
Towards Test-Time Continual Learning Agents (TTCL), NeurIPS 2026

**Student Volunteer:** ACS.

**Conference Reviewing:** NeurIPS, ICML, ICLR, AAAI, ACL, NAACL, ACS.

## Research Mentoring

**Anant Gupta:** M.S., Georgia Tech. 2025 – present.
**Karthik Singaravadivelan:** B.S., Georgia Tech. 2025 – present.
**Aryan Roy:** M.S., Georgia Tech. 2025 – present.
**Zihan Dong:** M.S., Georgia Tech. / Lenovo. 2024 – present.
**Ethan Haarer:** M.S., Georgia Tech. 2024 – 2025.

## Skills

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|---|---|
| **Programming** | Python, C/C++, CUDA, R, MATLAB, Java, C#, TEX, SQL |
| **ML Frameworks** | PyTorch, Hugging Face (Transformers, Diffusers, PEFT/LoRA, TRL), Accelerate, DeepSpeed, vLLM, JAX (working), Weights & Biases |
| **Infrastructure** | SLURM, Docker, multi-node A100/H100 clusters, Linux/Bash, Git |
| **Mathematics** | Information / probability theory, Bayesian inference, calculus, linear algebra, variational methods, score matching, optimization |
| **Research areas** | Continual / lifelong learning, LLM post-training (RLHF/PPO/DPO/meta-RL), diffusion & flow models, meta-learning, compositional generalization, concept formation |
| **Languages** | English (fluent), Cantonese (fluent), Mandarin Chinese (native) |

## Full Publication List

### Published and Accepted Papers

**Trust Region Continual Learning as an Implicit Meta-Learner** *2026*
**Zekun Wang**[*], Anant Gupta[*], Christopher J. MacLellan
*NeurIPS 2026*

**Diffusion Models and Concept Formation** *2026*
**Zekun Wang**, Karthik Singaravadivelan, Christopher J. MacLellan
*Advances in Cognitive Systems 2026*
**Ada Lovelace Award for Excellent Paper**

**A Rational Account of Categorization Based on Information Theory** *2026*
Christopher J. MacLellan, Karthik Singaravadivelan, Xin Lian, **Zekun Wang**, Pat Langley
*CogSci 2026*

**Decompose Sparsely Where You Should, Absorb Densely Where You Should Not** *2026*
Ruixuan Deng, Zehao Jin, **Zekun Wang**, Zihan Dong
*ICML 2026 Mechanistic Interpretability Workshop (Poster)*

**CobwebTM: Probabilistic Concept Formation for Lifelong and Hierarchical Topic Modeling** *2026*
Karthik Singaravadivelan[*], Anant Gupta[*], **Zekun Wang**, Christopher J. MacLellan
*Findings of ACL 2026*

**Grounded Concreteness: Human-Like Concreteness Sensitivity in Vision-Language Models** *2026*
Aryan Roy, **Zekun Wang**, Christopher J. MacLellan
*Findings of ACL 2026*

**Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion Models** *2026*
**Zekun Wang**[*], Anant Gupta[*], Zihan Dong, Christopher J. MacLellan
*ICLR 2026*

**Computer Vision Modeling of the Development of Geometric and Numerical Concepts in Humans** *2026*
**Zekun Wang**, Sashank Varma
*AAAI 2026*

**Deep Taxonomic Networks for Unsupervised Hierarchical Prototype Discovery** *2025*
**Zekun Wang**, Ethan Haarer, Tianyi Zhu, Zhiyi Dai, Christopher J. MacLellan
*NeurIPS 2025*

**Hierarchical Semantic Retrieval with Cobweb** *2025*
Anant Gupta[*], Karthik Singaravadivelan[*], **Zekun Wang**
*Advances in Cognitive Systems 12 (2025)*

**Efficient and Scalable Masked Word Prediction Using Concept Formation** *2025*
Xin Lian[*], **Zekun Wang**[*], Christopher J. MacLellan
*Cognitive Systems Research 92, 101371*

**Computer Vision Models Show Human-Like Sensitivity to Geometric and Topological Concepts** *2025*
**Zekun Wang**, Sashank Varma
*CogSci 2025*

**Taxonomic Networks: A Representation for Neuro-Symbolic Pairing** *2025*
**Zekun Wang**, Ethan L. Haarer, Nicki Barari, Christopher J. MacLellan
*NeuS 2025*

**Babysit A Language Model From Scratch: Interactive Language Learning by Trials and Demonstrations** *2025*
Ziqiao Ma*, **Zekun Wang***, Joyce Chai
*NAACL 2025*

**Has It All Been Solved? Open NLP Research Questions Not Solved by Large Language Models** *2024*
Oana Ignat*, Zhijing Jin*, Artem Abzaliev, Laura Biester, Santiago Castro, Naihao Deng, Xinyi Gao, Aylin Gunal, Jacky He, Ashkan Kazemi, Muhammad Khalifa, Namho Koh, Andrew Lee, Siyang Liu, Do June Min, Shinka Mori, Joan Nwatu, Veronica Perez-Rosas, Siqi Shen, **Zekun Wang**, Winston Wu, Rada Mihalcea
*LREC-COLING 2024*

**Foundations for AI-Assisted Formative Assessment Feedback for Short-Answer Tasks in Large-Enrollment Classes** *2022*
Susan E. Lloyd, Matthew D. Beckman, Dennis K. Pearl, Rebecca J. Passonneau, Zhaohui Li, **Zekun Wang**
*ICOTS 2022*

### Preprints

**VERSE: Verified Self-Evolving Optimizer for Agent Harnesses** *2026*
Zekai Wang, Yingqiang Ge, **Zekun Wang**, Hai Wang, Yuhui Xu, Joshua Frandsen, Shancong Fu, Ashia C. Wilson, Chandan K. Reddy
*arXiv preprint*

**Self-Consolidating Language Models** *2026*
**Zekun Wang***, Anant Gupta*, Zihan Dong, Christopher J. MacLellan
*Preprint*

**Test-Time Compositional Generalization in Diffusion Models via Concept Discovery** *2026*
**Zekun Wang**, Anant Gupta, Tianyi Zhu, Christopher J. MacLellan
*arXiv preprint*